

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 6_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Preethi is working on a project to automate sales tax calculations for items in a store. She wants to create a program that takes the price of an item and the sales tax rate as input and calculates the final price of the item after applying the sales tax.

Write a program using the class SalesTaxCalculator, which contains an overloaded method named calculateFinalPrice to handle both integer and double inputs. The program should also include a Main class that takes user input, calls the appropriate method from SalesTaxCalculator, and prints the final price of the item.

Formula Used: Final price = price + ((price * sales tax rate) / 100)

Input Format

The first line of input consists of an integer price (the price of the item for integer inputs).

The second line of input consists of an integer taxRate (the sales tax rate for integer inputs).

The third line of input consists of a double price (the price of the item for double inputs).

The fourth line of input consists of a double taxRate (the sales tax rate for double inputs).

Output Format

The first line of output prints an integer, representing the final price of the item after applying the sales tax for integer inputs (a and b).

The second line prints a double value, representing the final price of the item after applying the sales tax for double-value inputs (m and n), rounded to two decimal places.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 100

10

100.0

5.0

Output: 110

105.00

Answer

```
import java.util.Scanner;
```

```
// You are using Java
```

```
/**
```

```
 * A calculator class to compute the final price of an item after sales tax.
```

```
 * It uses method overloading to handle both integer and double data types.
```

```

*/
class SalesTaxCalculator {

    /**
     * Calculates the final price for integer inputs.
     * @param price The price of the item as an integer.
     * @param taxRate The sales tax rate as an integer percentage.
     * @return The final price as an integer.
     */
    public static int calculateFinalPrice(int price, int taxRate) {
        return price + ((price * taxRate) / 100);
    }

    /**
     * Calculates the final price for double inputs.
     * @param price The price of the item as a double.
     * @param taxRate The sales tax rate as a double percentage.
     * @return The final price as a double.
     */
    public static double calculateFinalPrice(double price, double taxRate) {
        return price + ((price * taxRate) / 100.0);
    }
}

/**
 * The main class to run the program, take user input, and display the results.
 */

class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int intPrice = scanner.nextInt();
        int intTaxRate = scanner.nextInt();
        double doublePrice = scanner.nextDouble();
        double doubleTaxRate = scanner.nextDouble();

        int finalPriceInt = SalesTaxCalculator.calculateFinalPrice(intPrice,
intTaxRate);
        double finalPriceDouble =
SalesTaxCalculator.calculateFinalPrice(doublePrice, doubleTaxRate);

```

```
        System.out.println(finalPriceInt);  
        System.out.format("%.2f", finalPriceDouble);  
    }  
}
```

Status : Correct

Marks : 10/10